Formalizing Co-Design

UBISS 2026 Workshop A

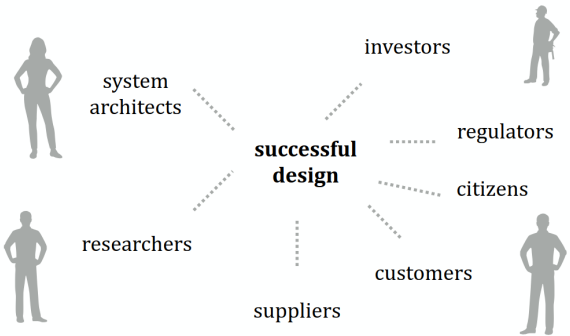
Alexandra Nilles Başak Sakçak Dylan Shell

Outline

- What is co-design?
- Approaches to co-design

What is Co-Design?

Collaborative Design



Source: Andrea Censi

Resources and Functionality

- Resources must be accounted for and assigned *costs*

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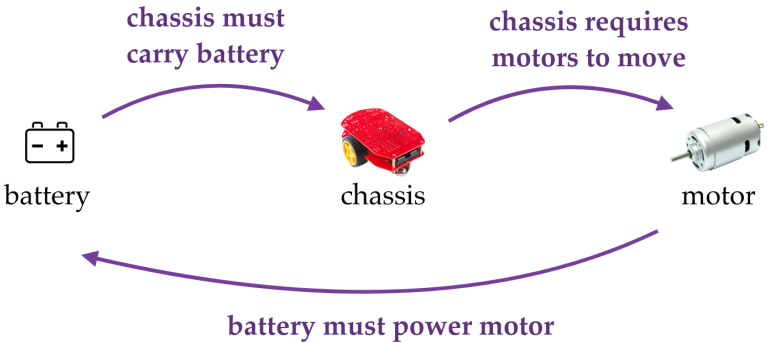
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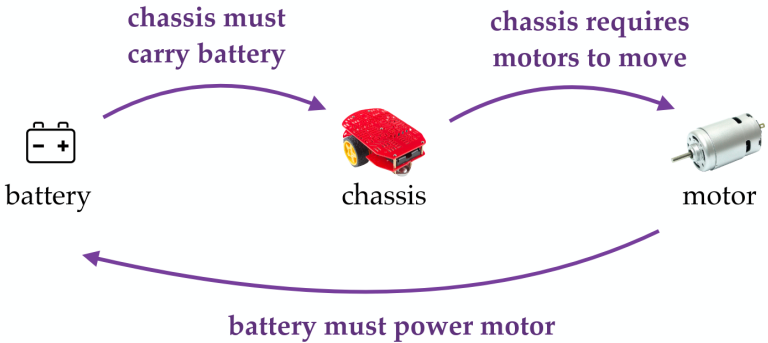
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- *Concurrent Design*

Feedback in Design

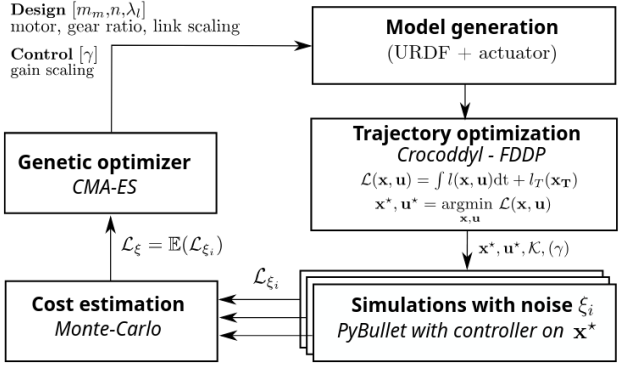


Feedback in Design



Irreducible complexity: either converges, or spirals into infinity.

Co-Design of Hardware and Control

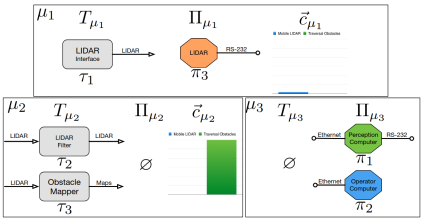


Fadini, Gabriele, Thomas Flayols, Andrea Del Prete, and Philippe Souères. "Simulation aided co-design for robust robot optimization." IEEE Robotics and Automation Letters 7, no. 4 (2022): 11306-11313.

Approaches to Solving Co-Design Problems

Optimization

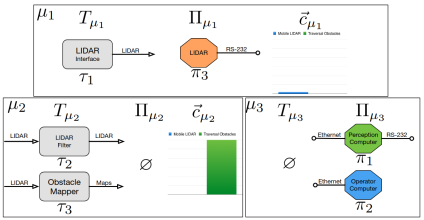
- Formulate modules with requirements and functionality as constraint functions



Ziglar, Jason, Ryan K. Williams, and Alfred Wicks. "Context-aware system synthesis, task assignment, and routing." Autonomous Robots 47.2 (2023): 193-210.

Optimization

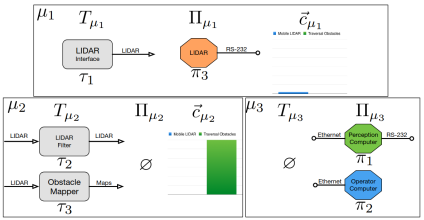
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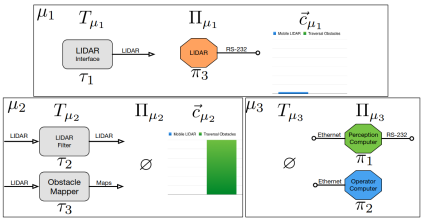
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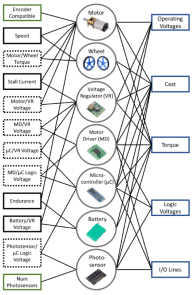
Optimization

- Formulate modules with requirements and functionality as constraint functions
- Ideally, use linear/quadratic programming
- Often NP-hard but "fast enough" with careful modelling
- Requires careful system approximations and simplifications



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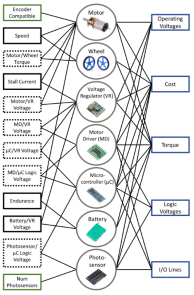
Combinatorial Optimization



Wilhelm, Andrew, and Nils Napp. "Constraint Programming for Component-Level Robot Design." In 2023 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS), pp. 460-466. IEEE, 2023.

- "the efficacy of constraint programming critically depends upon the orderings of variables and their domain values. . . "

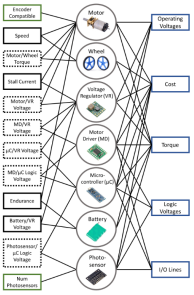
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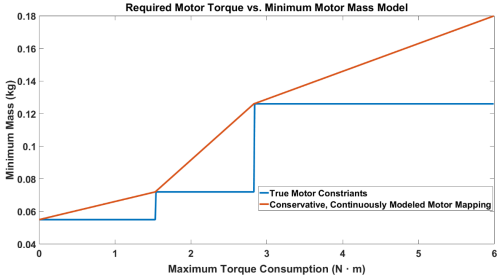


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- "the efficacy of constraint programming critically depends upon the orderings of variables and their domain values. . . "
- requires careful design of heuristics
- lots of opportunities for study of effective design processes!

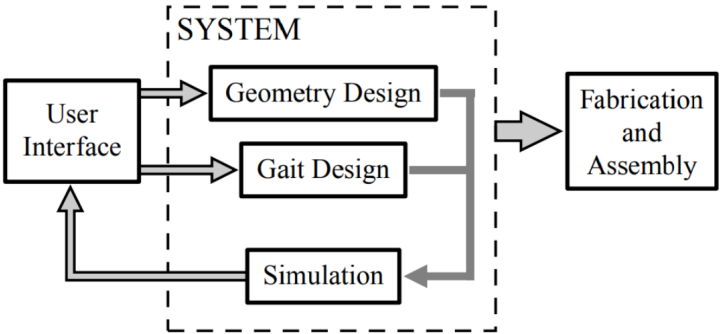
Gradient-Driven Optimization

- easier with “smooth” design spaces
- strategies for soft and rigid robots differ



Spielberg AE. Co-Optimization and Co-Learning Methods for Automated Design of Rigid and Soft Robots (Doctoral dissertation, Massachusetts Institute of Technology).

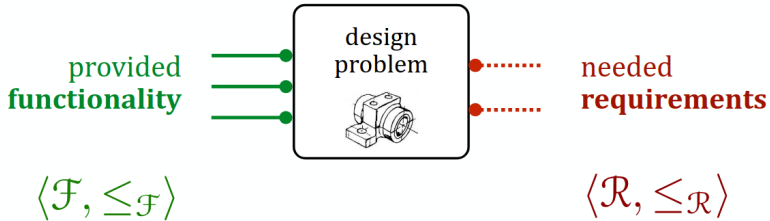
Simulation-Driven Co-Design



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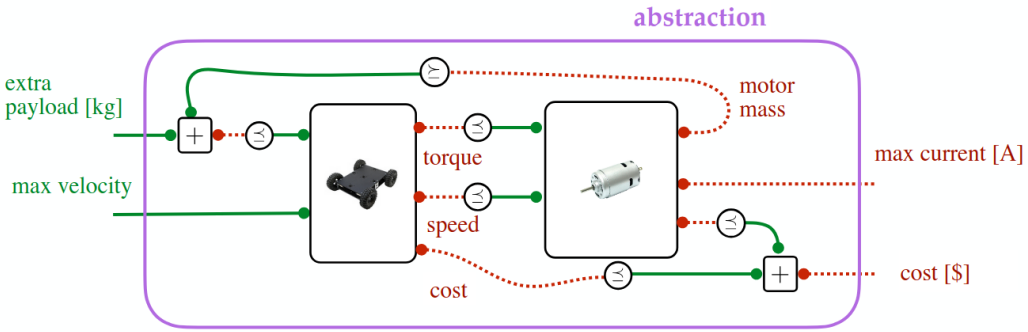
See also: genetic algorithms, reinforcement learning, etc.

Monotone Co-Design Problems (MCDPs)



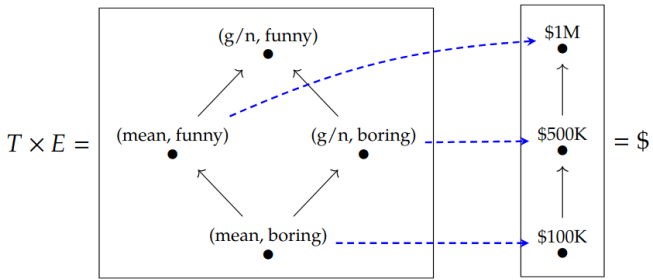
Source: Andrea Censi

Monotone Co-Design Problems (MCDPs)



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MCDPs and Category Theory

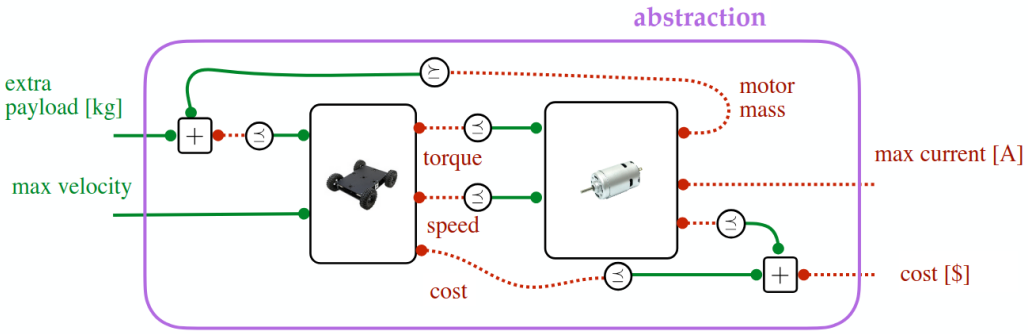


B. Fong, D.I. Spivak, An Invitation to Applied Category Theory, 2019. https://doi.org/10.1017/9781108668804.

Takeaway: Problem is well-structured (and fun). Known algorithms (Kleene's) can efficiently find solutions.

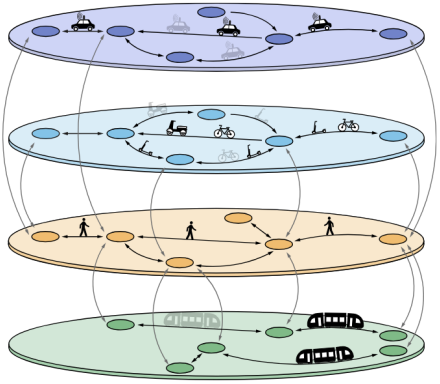
Monotone Assumption

When does it break down?



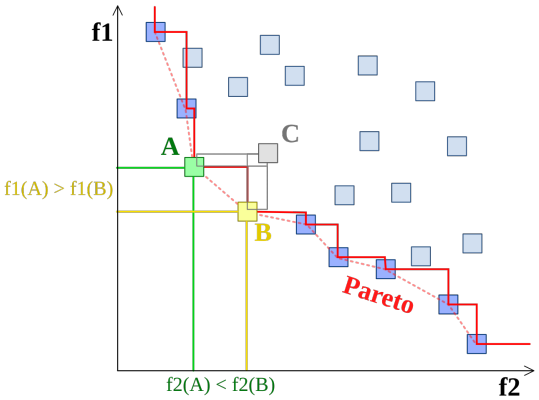
Source: Andrea Censi

Applications of MCDPs



Source: Zardini, Gioele, Nicolas Lanzetti, Andrea Censi, Emilio Frazzoli, and Marco Pavone. "Co-design to enable user-friendly tools to assess the impact of future mobility solutions." IEEE Transactions on Network Science and Engineering 10, no. 2 (2022): 827-844.

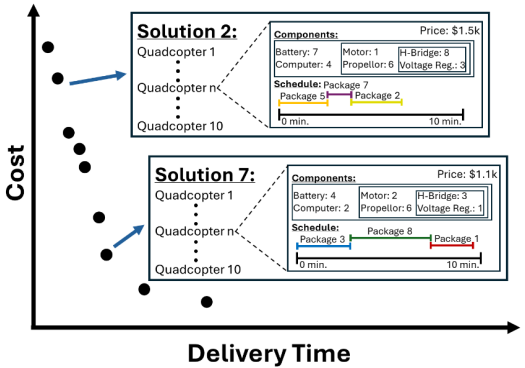
Solution Space of MCDPs



Source: Wikipedia, By Nojhan - Own work, CC BY-SA 3.0, https://commons.wikimedia.org/w/index.php?curid=770240

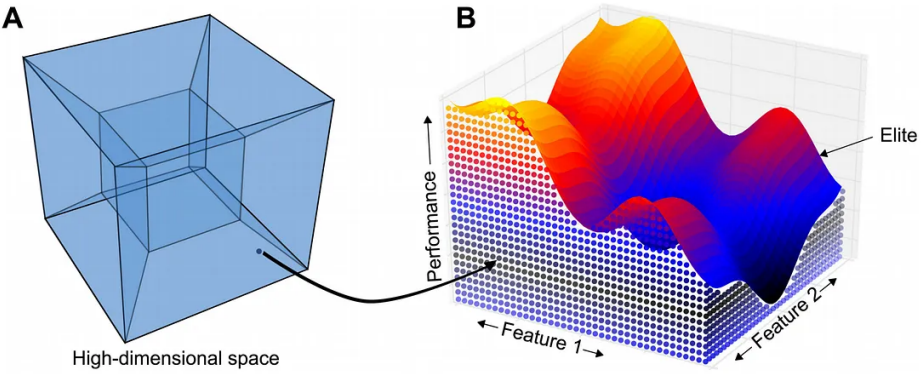
Pareto Front

Automatic Design of Monotone Subsystem Discretizations



Wilhelm, Andrew, and Nils Napp. "Monotone Subsystem Decomposition for Efficient Multi-Objective Robot Design." In 2025 IEEE International Conference on Robotics and Automation (ICRA), pp. 8114-8120. IEEE, 2025.

Quality-Diversity Algorithms

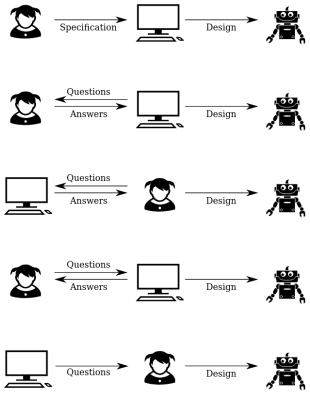


Mouret, Jean-Baptiste, and Jeff Clune. "Illuminating search spaces by mapping elites." arXiv preprint arXiv:1504.04909 (2015).

See also: Marc Toussaint keynote @ WAFR 2026

Human In-The-Loop

- Problem with prior approaches is you must fully model everything before you apply the algorithm
- "A lot of times, people don't know what they want until you show it to them." - Steve Jobs



Nilles, Alexandra Q., Dylan A. Shell, and Jason M. O'Kane. "Robot design: Formalisms, representations, and the role of the designer." arXiv preprint arXiv:1806.05157 (2018).

Summary

- Structure of robot design problem creates interesting theoretical questions in automated design

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- Traditional optimization approaches struggle with complexity of robot co-design problem
- There is no getting around careful encoding of resources, constraints, and functionality
- As systems scale, the importance of principled and high-assurance design methods becomes clear